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SEMICONDUCTOR DEVICE

Abstract

A semiconductor device includes first and second semiconductor layers of a first conductivity type, a third semiconductor layer of a second conductivity type, a plurality of electrodes, and a first insulating film. The second semiconductor layer is provided on the first semiconductor layer. The third semiconductor layer is provided on the second semiconductor layer with a first surface at a side opposite to the first semiconductor layer. The electrodes extend from the first surface into the second semiconductor layer. A first insulating film is provided between the second and third semiconductor layers and each of electrodes. The electrodes include first and second electrode groups. The first electrode group is arranged in one column in the first direction and apart from each other by a first distance. The first and second electrode groups are apart from each other by a second distance in the second direction.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2020-156337, filed on Sep. 17, 2020; the entire contents of which are incorporated herein by reference.

FIELD

[0002] Embodiments relate to a semiconductor device.

BACKGROUND

[0003] Semiconductor devices such as IGBTs (Insulated Gate Bipolar Transistors), diodes, and the like are used in power converters, for example, inverters and the like. A so-called FWD (Free Wheeling Diode), for example, is connected in anti-parallel with an IGBT. To increase the conversion efficiency, it is required for such a power converter to improve the characteristics of the FWD.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] FIGS. 1A and 1B are schematic views showing a semiconductor device according to an embodiment;

[0005] FIGS. 2A to 2C are schematic views showing characteristics of the semiconductor device according to the embodiment;

[0006] FIG. 3 is a graph showing another characteristic of the semiconductor device according to the embodiment;

[0007] FIGS. 4A and 4B are schematic views showing trench arrangements of the semiconductor device according to the embodiment;

[0008] FIGS. 5A and 5B are schematic views showing trench arrangements of semiconductor devices according to modifications of the embodiment;

[0009] FIGS. 6A and 6B are schematic views showing trench arrangements of semiconductor devices according to other modifications of the embodiment;

[0010] FIGS. 7A and 7B are schematic cross-sectional views showing semiconductor devices according to modifications of the embodiment;

[0011] FIG. 8 is a schematic view showing the configuration of an interconnect of a semiconductor device according to another modification of the embodiment;

[0012] FIGS. 9A and 9B are schematic views showing configurations of interconnects of a

semiconductor device according to an embodiment; and

[0013] FIG. **10** is a schematic view showing another example of the configuration of the interconnect in the semiconductor device according to the embodiment.

DETAILED DESCRIPTION

[0014] According to an embodiment, a semiconductor device includes first and second semiconductor layers of a first conductivity type, a third semiconductor layer of a second conductivity type, a plurality of electrodes, and a first insulating film. The second semiconductor layer is provided on the first semiconductor layer and includes a first-conductivity-type impurity with a lower concentration than a concentration of a first-conductivity-type impurity in the first semiconductor layer. The third semiconductor layer is provided on the second semiconductor layer. The third semiconductor layer includes a first surface at a side opposite to the second semiconductor layer. The first surface extends in a first direction and a second direction crossing the first direction. The electrodes are provided on the second semiconductor layer. The electrodes respectively extend inside trenches each having depths enough to extend from the first surface into the second semiconductor layer. A first insulating film is provided between the third semiconductor layer and one of the electrodes and between the second semiconductor layer and the one of the electrodes. The electrodes include first and second electrode groups. The first electrode group is arranged in one column in the first direction and apart from each other by a first distance. The second electrode group is arranged in another column in the first direction and apart from each other by the first distance. The first electrode group and the second electrode group are apart from each other by a second distance in the second direction, the first distance being greater than the second distance.

[0015] Embodiments will now be described with reference to the drawings. The same portions inside the drawings are marked with the same numerals; a detailed description is omitted as appropriate; and the different portions are described. The drawings are schematic or conceptual; and the relationships between the thicknesses and widths of portions, the proportions of sizes between portions, etc., are not necessarily the same as the actual values thereof. The dimensions and/or the proportions may be illustrated differently between the drawings, even in the case where the same portion is illustrated.

[0016] There are cases where the dispositions of the components are described using the directions of XYZ axes shown in the drawings. The X-axis, the Y-axis, and the Z-axis are orthogonal to each other. Hereinbelow, the directions of the X-axis, the Y-axis, and the Z-axis are described as an X-direction, a Y-direction, and a Z-direction. Also, there are cases where the Z-direction is described as upward and the direction opposite to the Z-direction is described as downward.

[0017] FIGS. **1A** and **1B** are schematic views showing a semiconductor device **1** according to an embodiment. FIG. **1A** is a cross-sectional view along the X-Z plane. FIG. **1B** is a plan view illustrating a cross section along line A-A shown in FIG. **1A**.

[0018] The semiconductor device **1** is, for example, a diode. The semiconductor device **1** is integrated and used with the IGBT. Embodiments described below are examples and are not limited thereto.

[0019] As shown in FIG. **1A**, the semiconductor device **1** includes a first semiconductor layer **11** of a first conductivity type, a second semiconductor layer **12** of the first conductivity type, and a third semiconductor layer **13** of the second conductivity type. The second semiconductor layer **12** is provided on the first semiconductor layer **11**. The third semiconductor layer **13** is provided on the second semiconductor layer **12**. The second semiconductor layer **12** includes a first-conductivity-type impurity with a lower concentration than the first-conductivity-type impurity in the first semiconductor layer **11**.

[0020] The first semiconductor layer **11** is, for example, an n-type cathode layer. The third semiconductor layer **13** is, for example, a p-type anode layer. In the following description, the first conductivity type is described as an n-type; and the second conductivity type is described as a p-

type.

[0021] The semiconductor device **1** further includes an electrode **30**. The electrode **30** is provided inside a trench AT that has a depth enough to extend from the surface of the third semiconductor layer **13** into the second semiconductor layer **12**. The electrode **30** extends in the third semiconductor layer **13** and the second semiconductor layer **12**. The electrode **30** is, for example, conductive polysilicon. For example, the electrode **30** is electrically insulated from the third and second semiconductor layers **13** and **12** by an insulating film **33**. The insulating film **33** is provided between the electrode **30** and the third semiconductor layer **13** and between the electrode **30** and the second semiconductor layer **12**. The insulating film **33** is, for example, a silicon oxide film.

[0022] The first semiconductor layer **11** is electrically connected to an electrode **10** (a cathode electrode). The third semiconductor layer **13** is electrically connected to an electrode **20** (an anode electrode). For example, the electrode **30** is electrically connected to the electrode **20** and has the same potential as the third semiconductor layer **13**. Embodiments are not limited thereto; for example, the electrode **30** may be electrically connected to another electrode that is not illustrated and may be biased to a different potential from the third semiconductor layer **13**.

[0023] As shown in FIG. **1B**, multiple electrodes **30** are provided respectively inside the multiple trenches AT. Some of the multiple electrodes **30** form, for example, a column in which electrodes **30** are arranged in a Y-direction; and multiple columns each including the electrodes **30** are arranged in the X-direction.

[0024] The multiple trenches AT, for example, are arranged so that the second semiconductor layer **12** includes a region positioned between neighboring trenches AT, and the region of the second semiconductor layer **12** is depleted when a prescribed reverse bias of, for example, 1 V is applied between the first semiconductor layer **11** and the third semiconductor layer **13**. In other words, a maximum width WM between the neighboring trenches AT is provided so that the current path between the neighboring trenches AT is pinched off.

[0025] For example, in a cross-section of the second semiconductor layer **12** parallel to X-Y plane, a distance between the neighboring trenches AT is defined as a length of a line segment between neighboring trenches AT when a straight line is drawn in the cross-section through any point in the second semiconductor layer **12**. The distance or spacing between the trenches AT, and the length of the trench AT are determined based on the position of the sidewall of the trench AT or the outer edge of the insulating film **33** included in the trench AT as a reference. The maximum distance between the neighboring trenches AT in the second semiconductor layer **12** is the maximum width WM.

[0026] FIGS. **2A** to **2C** and FIG. **3** are schematic views showing characteristics of the semiconductor device **1** according to the embodiment. FIGS. **2A** to **2C** are schematic views showing the cross-section from the surface to the trench bottom between the adjacent trenches AT. The depth of the trench AT is, for example, 5.5 μm . In FIGS. **2A** to **2C**, the depletion layers in the third and second semiconductor layers **13** and **12** under a prescribed reverse bias of, for example, 1 V are illustrated. The equipotential surfaces shown in the drawings illustrate the spreading of the depletion layer. FIG. **3** is a graph illustrating the reverse breakdown voltage of the semiconductor device **1**.

[0027] As shown in FIGS. **2A** to **2C**, the spacings between the adjacent trenches AT (referring to FIG. **1B**) are 1.6 micrometers (hereinbelow, μm), 2.0 μm , and 2.5 μm , respectively.

[0028] As shown in FIGS. **2A** and **2B**, the depleted portion of the second semiconductor layer **12** is provided between the adjacent trenches AT when the spacing between the adjacent trenches AT is 1.6 μm and 2.0 μm . Such a depleted state is so called pinch-off. As shown in FIG. **2C**, a portion of the second semiconductor layer **12** is not depleted between the adjacent trenches AT when the spacing between the adjacent trenches AT is 2.5 μm .

[0029] In FIG. **3**, illustrated is the relationship between the spacing between the adjacent trenches AT and the reverse breakdown voltage (the breakdown voltage). As shown in FIG. **3**, the

breakdown voltage decreases as WM increases.

[0030] According to the embodiment, the semiconductor device **1** includes the electrode **30** that has a trench structure to ensure the reverse breakdown voltage and the anode-cathode breakdown immunity. However, by providing the electrode **30**, the surface area in the X-Y plane of the second semiconductor layer **12** is reduced. Compared to the case in which the electrode **30** is not provided, the conduction path of the carriers is narrow, and the carrier density (the density of the electrons and holes) in the third semiconductor layer **13** is high.

[0031] For example, in a power conversion device such as an inverter, etc., it is desirable to improve the characteristics such as the on-voltage, the recovery time, the safe operation region in recovery, and the current and voltage oscillation in recovery, etc. Here, the on-voltage is the voltage drop in the conducting state; the recovery time is the extinction time of the recovery current in reverse recovery; and the safe operation region in recovery is an operation region in which breakdown does not occur even when a voltage is applied while a reverse recovery current flows. Among these characteristics, it is important to widen the safe operation region in recovery while shortening the recovery time.

[0032] In the semiconductor device **1**, by providing a plurality of the trenches AT with a regulated arrangement, in which the electrodes **30** are provided respectively, a uniform avalanche phenomenon can be generated at the bottoms of the trenches AT; and the breakdown immunity may be increased. Moreover, the reverse breakdown voltage may be increased by providing the spacing WM between the adjacent trenches AT that is less than the spacing at which pinch-off occurs even if the low reverse bias, for example, about 1 V is applied.

[0033] By forming the trenches AT, however, the carrier density in the second and third semiconductor layers **12** and **13** is increased in the on-state. Therefore, the recovery characteristics degrade while transitioning from the on-state to the off-state, and the recovery loss is large compared to when the trenches AT are not provided.

[0034] In the semiconductor device **1**, the trench AT is provided with a less occupancy ratio in the X-Y cross section of the third semiconductor layer **13** so that the density of the holes injected from the third semiconductor layer **13** into the second semiconductor layer **12** is reduced. Thereby, the recovery loss may be reduced while suppressing the reduction of the reverse breakdown voltage and the breakdown immunity.

[0035] FIGS. **4A** and **4B** are schematic plan views illustrating configurations of the semiconductor device **1** according to the embodiment. FIGS. **4A** and **4B** are plan views illustrating arrangements of the trench AT. The electrode **30**, the insulating film **33**, and the third semiconductor layer **13** are not illustrated in the layout drawings of the trenches AT shown below.

[0036] As shown in FIG. **4A**, for example, the trench AT has a rectangular X-Y cross section in which sides extend in the X-direction and the Y-direction. The multiple trenches AT are arranged in a matrix configuration by being arranged in the Y-direction to form columns and by being arranged in the X-direction to form rows. A length LY in the Y-direction of the trench AT is greater than a length LX in the X-direction. The adjacent trenches AT in the X-direction are arranged with a spacing WDX. The adjacent trenches AT in the Y-direction are arranged with a spacing WDY. The spacing between the adjacent trenches AT has a maximum, for example, in the diagonal direction of the arrangement of the multiple trenches AT. Here, the diagonal direction is the direction directed from a coordinate (LX, LY) toward another coordinate (LX+WDX, LY+WDY) in the X-Y plane. The maximum spacing WM is, for example, the distance between the end portions of the neighboring trenches AT in the diagonal direction and is not more than 2 μm .

[0037] In the example shown in FIG. **4B**, first and second columns each include the trenches AT of the multiple trenches AT arranged in the Y-direction. The first column is adjacent to the second column in the X-direction. Each trench AT in the first column and a space between the trenches AT in the second column are arranged in the X-direction. The spacing WDY between the trenches AT that are adjacent in the Y-direction is less than the length LY in the Y-direction of the trench AT.

The spacing WDX between the trenches AT that are adjacent in the X-direction is, for example, not more than 1 μm . In such a case, the spacing WM between the adjacent trenches AT has a maximum between the trenches AT that are adjacent in the Y-direction. The spacing WM between the adjacent trenches AT is, for example, not more than 2 μm .

[0038] FIGS. 5A and 5B are schematic views illustrating configurations of the semiconductor device **1** according to modifications of the embodiment. FIGS. 5A and 5B are plan views illustrating arrangements of the trenches AT.

[0039] As shown in FIG. 5A, the first and second columns each includes the trenches AT of the multiple trenches AT that are arranged in the Y-direction. The second column is adjacent to the first column. Each trench AT in the first column and a space between the trenches AT in the second column are arranged in the X-direction. The spacing WDY between the trenches AT that are adjacent in the Y-direction is greater than the length LY in the Y-direction of each trench AT. WDX is, for example, not more than 1 μm . In such a case as well, the spacing WM between the adjacent trenches AT has a maximum between the trenches AT that are adjacent in the Y-direction. The spacing WM between the adjacent trenches AT is, for example, not more than 2 μm .

[0040] As shown in FIG. 5B, for example, the trench AT may have a circular cross section in the X-Y plane. The trenches AT are arranged in, for example, the diagonal direction of the arrangement thereof. The spacing WM is provided between the trenches AT that are adjacent in the diagonal direction, and is, for example, not more than 2 μm .

[0041] FIGS. 6A and 6B are schematic views showing configurations of the semiconductor device **1** according to other modifications of the embodiment. FIGS. 6A and 6B are plan views illustrating the arrangements of the trenches AT.

[0042] As shown in FIG. 6A, the trench AT may have, for example, a circular-ring-shaped cross section in the X-Y plane. The trench AT surrounds a portion of the third semiconductor layer **13** and a portion of the second semiconductor layer **12**. The trenches AT are arranged in, for example, the diagonal direction of the arrangement thereof. The spacing WM is provided between the trenches AT that are adjacent in the diagonal direction, and is, for example, not more than 2 μm . The inner diameter of the trench AT is, for example, not more than 2 μm .

[0043] As shown in FIG. 6B, the trench AT has, for example, a hexagonal-ring-shaped cross section in the X-Y plane. The trenches AT are arranged in, for example, the diagonal direction of the arrangement thereof. The spacing WM is provided between the trenches AT that are adjacent in the diagonal direction, and is, for example, not more than 2 μm . The trench inner diameter is, for example, not more than 2 μm . The exterior shape of the cross section of the trench AT is not limited to hexagonal and may be another polygon.

[0044] In the arrangement of the trenches AT described above, it is preferable for the trench AT to have the total cross-sectional area in the X-Y plane, for example, less than the surface area of the third semiconductor layer **13** in the X-Y plane. That is, the third semiconductor layer has a surface area greater in the cross-section than a sum of a cross-sectional area of the plurality of first insulating films and a cross-sectional area of the plurality of electrodes at a level same as a level of the cross-section. Thereby, the density of the holes injected from the third semiconductor layer **13** into the second semiconductor layer **12** can be reduced, and the carrier density in the second semiconductor layer **12** can be reduced.

[0045] FIGS. 7A and 7B are schematic views showing semiconductor devices **2** and **3** according to modifications of the embodiment.

[0046] As shown in FIG. 7A, the semiconductor device **2** includes the trench AT that has a reverse-tapered cross-sectional shape in the X-Z plane. The trench AT of the semiconductor device **2** is provided with the shape such that the bottom of the trench AT positioned in the second semiconductor layer **12** has the width in the X-direction greater than a width in the X-direction at a portion thereof that is positioned in the third semiconductor layer **13**. The electrode **30** that is provided inside the trench AT may have a similar cross-sectional shape in the X-Z plane.

[0047] In the semiconductor device **2**, the spacing WM in the X-direction is ensured between the bottom surfaces of the adjacent trenches AT. The spacing WM is provided so that the pinch-off occurs even if the low reverse bias, for example, 1 V is applied. In contrast, the width in the X-direction of the third semiconductor layer **13** is greater than the spacing WM between the bottom surfaces of the trenches AT.

[0048] As shown in FIG. 7B, the semiconductor device **3** includes the trench AT that has the width in the X-direction at the portion positioned in the second semiconductor layer **12** greater than the width in the X-direction at another portion thereof that is positioned in the upper portion of the third semiconductor layer **13**. The electrode **30** that is provided inside the trench AT also has a similar cross-sectional shape in the X-Z plane.

[0049] In the semiconductor device **3**, the spacing WM in the X-direction is ensured between the portions of the adjacent trenches AT positioned in the second semiconductor layer **12**. The spacing WM is provided so that the pinch-off occurs under a reverse bias. In contrast, the width in the X-direction of the upper portion of the third semiconductor layer **13** is greater than the spacing WM.

[0050] Thus, in the semiconductor devices **2** and **3**, the multiple trenches AT are arranged so that the spacing WM between the bottoms of the adjacent trenches AT is less than the pinch-off spacing. The spacing of the upper portion of the third semiconductor layer **13** is greater than WM, and thus, the carrier density can be reduced by increasing the occupancy ratio of the third semiconductor layer **13** in the X-Y plane.

[0051] FIG. 8 is a schematic view showing the configuration of an interconnect **40** of a semiconductor device **4** according to an embodiment.

[0052] In the semiconductor device **4**, similarly to the semiconductor device **1** shown in FIGS. 1A and 1B, the third semiconductor layer **13** and the electrode **30** are electrically connected by the interconnect **40**. The interconnect **40** is, for example, a metal film that is provided on the third semiconductor layer **13**, the electrode **30**, and the insulating film **33**. The interconnect **40** is connected to the anode electrode **20**. In any of the semiconductor devices according to embodiments, the interconnect **40** may be provided in the manner shown in FIG. 8.

[0053] FIGS. 9A and 9B are schematic views showing configurations of interconnects of a semiconductor device **5** according to an embodiment.

[0054] According to the embodiment, the semiconductor device **5** may be operate under different potentials applied respectively to the third semiconductor layer **13** and the electrode **30**. In addition to the configuration of the semiconductor device **1**, the semiconductor device **5** further includes an insulating film **50**, an interconnect **40A**, and an interconnect **40B**. FIG. 9A is a schematic plan view illustrating a semiconductor device provided with the interconnects **40A** and **40B** that are added to the configuration shown in FIG. 4A. FIG. 9B is a cross-sectional view along line B-B shown in FIG. 9A.

[0055] The insulating film **50** is provided on the third semiconductor layer **13**, the electrode **30**, and the insulating film **33**. The interconnect **40A** (shown by the single dot-dash lines in FIG. 9A) extends in the X-direction on the insulating film **50**. The multiple interconnects **40A** are connected with an electrode **20A** that extends in the Y-direction.

[0056] The interconnect **40A** is connected to the multiple electrodes **30** via contacts **30c** (shown by diagonal-line portions in FIG. 9A) that extend through the insulating film **50**. The interconnect **40A** is electrically isolated from the third semiconductor layer **13**.

[0057] The interconnect **40B** extends in the X-direction on the insulating film **50**. The interconnect **40B** (shown by the broken lines in FIG. 9A) is connected to the third semiconductor layer **13** via a contact **13c** that extends through the insulating film **50**. The multiple interconnects **40B** are connected with an electrode **20B** that extends in the Y-direction. The interconnect **40B** is electrically isolated from the electrode **30**. The interconnect **40A** and the interconnect **40B** are apart from each other in the Y-direction and are electrically isolated. The interconnect **40A** and the interconnect **40B** are alternately arranged in the Y-direction. The interconnect **40A** and the

interconnect **40B** are, for example, metal films.

[0058] FIG. **10** is a schematic view showing another example of the configuration of the interconnect **40** of the semiconductor device **5** according to the embodiment. FIG. **10** is a schematic plan view showing an example of the interconnects **40A** and **40B** provided in the semiconductor device that has the configuration shown in FIG. **4B**. The contacts of the interconnect **40B** are not illustrated in FIG. **10**.

[0059] As shown in FIG. **10**, by providing the contacts **30c** of the interconnect **40A** at the end portions in the Y-direction of the trenches AT, the ratio of the contact area to the area occupied by the interconnect **40A** becomes large compared to a case in which the contacts **30c** are provided at the centers of the trenches AT.

[0060] While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel embodiments described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the invention.

Claims

1-14. (canceled)

15. A semiconductor device, comprising: a first semiconductor layer of a first conductivity type; a second semiconductor layer of the first conductivity type provided on the first semiconductor layer, the second semiconductor layer including a first-conductivity-type impurity with a lower concentration than a concentration of a first-conductivity-type impurity in the first semiconductor layer; a third semiconductor layer of a second conductivity type provided on the second semiconductor layer, the third semiconductor layer including a first surface at a side opposite to the second semiconductor layer, the first surface extending in a first direction and a second direction crossing the first direction; a plurality of electrodes provided on the second semiconductor layer, the plurality of electrodes respectively extending inside trenches, the trenches each having depths enough to extend from the first surface into the second semiconductor layer; a first insulating film provided between the third semiconductor layer and one of the plurality of electrodes and between the second semiconductor layer and the one of the plurality of electrodes; a second electrode electrically connected to the first semiconductor layer; a second insulating film provided on the first surface; a first interconnect provided above the first surface, the first interconnect being electrically connected with the third semiconductor layer; and a plurality of second interconnects provided on the second insulating film, the second interconnects being electrically connected to the plurality of electrodes, the second interconnects being apart respectively from the first interconnect, wherein the semiconductor device is a diode.

16. The device according to claim 15, wherein a first electrode group of the plurality of electrodes is arranged in one column in the first direction and apart from each other by a first distance, a second electrode group of the plurality of electrodes is arranged in another column in the first direction and apart from each other by the first distance, and the first electrode group and the second electrode group are apart from each other by a second distance in the second direction.

17. The device according to claim 16, wherein the first interconnect includes a plurality of portions that extends in the second direction, the plurality of second interconnects extends in the second direction, and the second interconnects are apart respectively from the portions of the first interconnect in the first direction.

18. The device according to claim 16, wherein the first distance is greater than the second distance.

19. The device according to claim 15, wherein each of the trenches has a first length in the first

direction, and a second length in the second direction, and the second length is less than the first length.

20. The device according to claim 16, wherein each of the trenches has a first length in the first direction, and a second length in the second direction, the second length is less than the first length, and the first distance is less than the first length in the first and second electrode groups.

21. The device according to claim 16, wherein the first distance is provided such that a portion of the second semiconductor layer between two adjacent electrodes of the plurality of electrodes is depleted when a prescribed voltage is applied between the first semiconductor layer and the third semiconductor layer.

22. The device according to claim 15, wherein a current path of the second semiconductor layer positioned between the plurality of electrodes is pinched off when a prescribed voltage is applied between the first semiconductor layer and the third semiconductor layer and between the first semiconductor layer and the plurality of electrodes.

23. The device according to claim 16, wherein the first distance is not more than 2 micrometers.

24. The device according to claim 15, wherein a surface area of the third semiconductor layer at the first surface is greater than a sum of a surface area of the first insulating film and a surface area of the plurality of electrodes at a level same as a level of the first surface.

25. The device according to claim 16, wherein the first electrode group and the second electrode group are adjacent in the second direction, and each electrode of the second electrode group and a space between adjacent electrodes of the first electrode group are arranged in the second direction.

26. The device according to claim 16, wherein the electrodes of the first and second electrode groups are apart from each other in the first direction by a third distance in the third semiconductor layer, the third distance being less than the first distance.

27. The device according to claim 15, wherein the trench has a circular shape in a cross-section parallel to the first surface.

28. The device according to claim 15, wherein the trench has a polygonal shape in a cross-section parallel to the first surface.

29. The device according to claim 15, wherein the trenches each surround portions of the second semiconductor layer and the third semiconductor layer.

30. The device according to claim 15, wherein a first spacing in the second semiconductor layer between adjacent trenches is less than a second spacing in the third semiconductor layer between the adjacent trenches.

31. The device according to claim 15, wherein no semiconductor layer of the first conductivity type is provided on the third semiconductor layer.
